Reproducibility and Good Practices in Science

Pietro Franceschi | pietro.franceschi@fmach.it

Why does this matter?

Science is the best tool we have to understand the natural world.

Why does this matter?

Science is the best tool we have to understand the natural world.

But science is done by **humans**, and humans make mistakes — or worse.

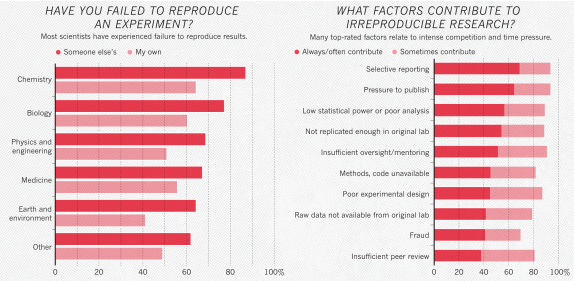
Why does this matter?

Science is the best tool we have to understand the natural world.

But science is done by **humans**, and humans make mistakes — or worse.

The question is not *whether* errors happen, but **whether we can detect and correct them**.

The Scientific Method



Baker, M. (2016). 1,500 scientists lift the lid on reproducibility. *Nature*, 533, 452–454. Survey of **1,576 researchers**.

The cornerstone of science is **reproducibility**: an independent group must be able to repeat your experiment and obtain the same result.

What can go wrong?

- **Poor experimental design** $\rightarrow$ systematic bias

What can go wrong?

- **Poor experimental design** $\rightarrow$ systematic bias
- **Inadequate statistical analysis** $\rightarrow$ spurious findings

What can go wrong?

- **Poor experimental design** $\rightarrow$ systematic bias
- **Inadequate statistical analysis** $\rightarrow$ spurious findings
- **Data analysis errors** $\rightarrow$ wrong conclusions from correct data

What can go wrong?

- **Poor experimental design** $\rightarrow$ systematic bias
- **Inadequate statistical analysis** $\rightarrow$ spurious findings
- **Data analysis errors** $\rightarrow$ wrong conclusions from correct data
- **Confirmation bias** $\rightarrow$ seeing what we want to see

What can go wrong?

- **Poor experimental design** $\rightarrow$ systematic bias
- **Inadequate statistical analysis** $\rightarrow$ spurious findings
- **Data analysis errors** $\rightarrow$ wrong conclusions from correct data
- **Confirmation bias** $\rightarrow$ seeing what we want to see
- **Fraud and misconduct** $\rightarrow$ active manipulation

A spectrum of failure

The "reproducibility crisis"

A large fraction of published scientific results cannot be reproduced by independent groups. Estimates range from **50% to 70%** of findings in some fields.

We will look at **three real cases** spanning different failure modes:

1. Honest delusion — Benveniste & homeopathy (Nature, 1988)
2. Fatal errors — The Potti scandal at Duke University
3. Correlation is not causation — Chocolate and Nobel Prizes (NEJM, 2012)

Section 1

Case 1: The Benveniste Affair

The original claim

In **June 1988**, *Nature* published a paper by Dayenas, Beauvais, and **J. Benveniste** from INSERM 200, Paris.

The claim: human basophils degranulate in response to anti-IgE antibodies diluted to $\mathbf{10^{-120}}$ — far beyond Avogadro's limit.

Not a single antibody molecule could be present in the solution.



816 SCIENTIFIC PAPER NATURE VOL. 333 30 JUNE 1988

Human basophil degranulation triggered by very dilute antiserum against IgE

E. Dayenas, F. Beauvais, J. Amara*, M. Oberbaum†, B. Robinzon ‡, A. Miadonna §, A. Tedeschi, B. Pomeranz‖, P. Fortner‖, P. Belon, J. Sainte-Laudy, B. Poitevin & J. Benveniste‖

INSERM U. 200, Universite Paris-Sud, 52 rue des Carnets, 92140 Clamart, France
* Rath Res Art Institute of Clinical Immunology, Kaplan Hospital, Rehovot 76100, Israel
† Department of Animal Sciences, Faculty of Agriculture, PO Box 12, The Hebrew University of Jerusalem, Rehovot 76100, Israel
‡ Department of Internal Medicine, Infectious Diseases and Immunopathology, University of Milano, Ospedale Maggiore Policlinico, Milano, Italy
§ Departments of Zoology and Physiology,* Ramsay Wright Zoological Laboratories, University of Toronto, 25 Harbord Street, Toronto, Ontario M5S 1A1, Canada
‖ To whom correspondence should be addressed.

When, human polymorphonuclear basophils, a type of white blood cell with antibodies of the immunoglobulin E (IgE) type on its surface, are exposed to anti-IgE antibodies, they release histamine from their intracellular granules and change their staining properties. The latter can be demonstrated at dilutions of anti-IgE that range from 1×10^2 to 1×10^{120}; over that range, there are successive peaks of degranulation from 40 to 60% of the basophils, despite the calculated absence of any anti-IgE molecules at the highest dilutions. Since dilutions need to be accompanied by vigorous shaking for the effects to be observed, transmission of the biological information could be related to the molecular organization of water.

THE antibodies responsible for human immediate hypersensitivity belong to the IgE isotype[1]. The most salient feature of IgE is its capacity to bind to mast cell and polymorphonuclear basophil membranes through receptors with high affinity[2]. Human basophils are specifically challenged by immunological stimuli such as allergens or anti-IgE antiserum that can bridge IgE molecules at the membrane[3]. This interaction initiates transmembrane and intracellular signals followed by granule exocytosis with the release of histamine and loss of metachromatic staining of basophil granules by a basic dye such as toluidine blue. Optical basophil degranulation is well correlated with other *in vitro* and *in vivo* procedures for the diagnosis of allergy[4].

In preliminary experiments, degranulation of human basophils contained in leukocyte suspensions was induced not only by the usual concentration of anti-IgE antibody (1 x 10^2 dilution of anti-IgE antiserum, corresponding to 2.2 x 10^{-9} M anti-IgE antibody in the assay), but also by very low concentrations of this antibody (2.2 x 10^{-109} M), where the number of IgG anti-IgE molecules in the assay is supposedly too low to trigger the process. We therefore explored this phenomenon.

Human IgE was purified from a pool of sera[5]. Rb(Fc) antiserum (1 mg specific antibody per ml) was prepared in HEPES-buffered Tyrode's solution containing human serum albumin (USA) down to 1 x 10^{-2} dilution, corresponding to a 2.2 x 10^{-9} M theoretical concentration (th) in the assay (see Fig. 1 legend for methods). The expected basophil degranulation, which was assessed by counting cells with metachromatical properties, was observed after exposure of leukocyte preparations to low antiserum dilutions with a maximum at ~ 1 x 10^{-2} dilution. Successive peaks of degranulation varying between 40 and 60% more than found down to 1 x 10^{-6} dilution, with periods of 6 to 9 tenfold dilutions (Fig. 1a). In other experiments, the antiserum was serially diluted a hundred-fold down to 1 x 10^{-20} (to give 2.2 x 10^{-29} M th in the assay) and similar results were obtained (Fig. 1b). Degranulation induced by high dilutions of the antiserum was suppressed by the addition of anti-Rb(Fc) at a range of dilutions down to 1 x 10^{6}, when at least 50 similar results were obtained at one or the other part of the high dilution scale in the participating laboratories (Toronto, preliminary results). As controls, goat antihuman IgG (Fc) antiserum (Fig. 1b, e = 4) or Tyrode's solution containing HSA (e = 5) were diluted down to 1 x 10^{-10} and 1 x 10^{6}, respectively. Cells incubated in conditions identical to those with anti-IgE antiserum gave no significant degranulation. The repetitive waves of anti-IgE-induced degranulation were reproducible, but the peaks of degranulation could shift by one or two dilutions with every fresh sequential dilution of anti-IgE and depended on the blood sample. The waves of basophil degranulation were also obtained with monoclonal anti-IgE (Fig. 1c) and with sera from allergic individuals, such as monoclonal anti-human IgE antibodies, specific antigen in allergic patients or in peroxidase-immunized rabbits, phosphilipase A2 from bee venom or porcine pancreas, the Na^+ ionophore monensin (up to 90% degranulation at 1 x 10^{-6} M th) and the Ca^{++} ionophores A23187 and mannequin (1 x 10^{-6} M th). The specificity of the observed effects at high dilutions (already noted when comparing antiserum against IgE with antiserum against IgG) was further strikingly illustrated in the ionophore experiments, because removing the corresponding ion from the cellular environment blunted basophil

Table 1 Basophil counts after exposure to anti-IgE antiserum at low and high dilutions

Samples	Experiment 1	Experiment 2	Experiment 3	Experiment 4
Tyrode's-HSA*	81.3±1.2†	89.0±3.3	81.7±2.2	106.7±1.4
Tyrode's-HSA	80.0±1.5	88.0±2.5	82.7±1.5	107.7±0.9
Tyrode's-HSA	80.0±1.5	88.0±2.5	81.7±1.0	105.7±0.9
rIgE 1 × 10^{-6}	37.3±1.8 (56)	42.3±4.4 (51)	37.7±0.7 (66)	49.0±1.5 (62)
rIgE 1 × 10^{-7}	37.3±1.8 (56)	42.3±4.4 (51)	37.7±0.7 (66)	49.0±1.5 (62)
rIgE 1 × 10^{-8}	76.0±1.1 (4)	89.7±1.8 (-1)	77.7±1.8 (4)	74.7±10.0 (31)
rIgE 1 × 10^{-9}	55.0±1.4 (33)	52.7±1.4 (41)	36.0±0.6 (57)	48.3±2.4 (55)
rIgE 1 × 10^{-5}	45.0±0.5 (44)	35.0±1.0 (61)	41.3±2.3 (49)	49.3±2.3 (54)
rIgE 1 × 10^{-4}	49.0±2.7 (40)	30.3±0.3 (64)	51.0±2.1 (32)	74.3±2.3 (31)
rIgE 1 × 10^{-3}	79.0±2.3 (2)	85.0±0.5 (5)	79.3±1.7 (3)	105.3±0.7 (1)

Blind experiments: test tubes were randomly coded twice by two independent pairs of observers and assayed. The codes were simultaneously broken at the end of all experiments. Cells were fixed in 2% glutaraldehyde and stained with 0.1% toluidine blue.
* Decoded additional tubes for negative (Tyrode's-HSA) or positive (rIgE 1 x 10^{-6}) controls. ‡ Data represent the mean ± s.e. of basophil number actually counted in triplicate (see legend to Fig. 1 for methods). § Number in parenthesis indicates percentage degranulation compared with Tyrode's-HSA.

What does it mean?

If anti-IgE retains its biological effectiveness at 10^{-120} dilution...

What does it mean?

If anti-IgE retains its biological effectiveness at 10^{-120} dilution...

...then **water has memory** of the molecules once dissolved in it.

What does it mean?

If anti-IgE retains its biological effectiveness at 10^{-120} dilution...

...then **water has memory** of the molecules once dissolved in it.

This would be the **scientific basis for homeopathy**.

What does it mean?

If anti-IgE retains its biological effectiveness at 10^{-120} dilution...

...then **water has memory** of the molecules once dissolved in it.

This would be the **scientific basis for homeopathy**.

Nature published it **anyway** — with an unprecedented *editorial reservation*, and on the condition that an independent team could investigate the laboratory.

Nature sends an investigation team

The team that visited INSERM 200 in July 1988:

- **John Maddox** — Editor of *Nature*

Nature sends an investigation team

The team that visited INSERM 200 in July 1988:

- **John Maddox** — Editor of *Nature*
- **James Randi** — Professional magician and skeptic (MacArthur Fellow)

Nature sends an investigation team

The team that visited INSERM 200 in July 1988:

- **John Maddox** — Editor of *Nature*
- **James Randi** — Professional magician and skeptic (MacArthur Fellow)
- **Walter W. Stewart** — Expert in scientific fraud and misconduct (NIH)

Nature sends an investigation team

The team that visited INSERM 200 in July 1988:

- **John Maddox** — Editor of *Nature*
- **James Randi** — Professional magician and skeptic (MacArthur Fellow)
- **Walter W. Stewart** — Expert in scientific fraud and misconduct (NIH)

Nature sends an investigation team

The team that visited INSERM 200 in July 1988:

- **John Maddox** — Editor of *Nature*
- **James Randi** — Professional magician and skeptic (MacArthur Fellow)
- **Walter W. Stewart** — Expert in scientific fraud and misconduct (NIH)

One week on-site. Access to **laboratory notebooks**. They imposed **strictly blinded** protocols.

The result: a delusion

NATURE VOL. 334 28 JULY 1988 NEWS AND VIEWS 287

"High-dilution" experiments a delusion

The now celebrated report by Dr J. Benveniste and colleagues elsewhere is found, by a visiting Nature *team, to be an insubstantial basis for the claims made for them.*

Two remarkable claims made in *Nature* (**333**, 816; 1988) by Dr Jacques Benveniste and his associates are based chiefly on an extensive series of experiments which are statistically ill-controlled, from which no independent effort has been made to exclude significant error, including experimenter bias, and whose interpretation has been clouded by the exclusion of measurements in conflict with the claim that anti-IgE at "high dilution" will degranulate basophils. The phenomenon described is not reproducible in the ordinary meaning of that word.

We conclude that there is no substantial basis for the claim that anti-IgE at high dilution (in the sense intended) retains its biological effectiveness, and that the hypothesis that water can be imprinted with the memory of past solutes is unnecessary as it is fanciful.

We use the term "high dilution" reluctantly; these solutions contain no molecules of anti-IgE, and so are not solutions in the ordinary sense. "Solute-free solutions" would be more accurate.

Our conclusion is based on a week-long visit to Dr Benveniste's laboratory, the INSERM unit for immunopharmacology and allergy (otherwise INSERM 200) at Clamart, in the western suburbs of Paris, during the week beginning 4 July. Among other things, we were dismayed to learn that the salaries of two of Dr Benveniste's coauthors of the published article are paid for under a contract between INSERM and Dr Jacques Benveniste Laboratoire Cie., a supplier of pharmaceuticals and homoeopathic medicines, as were our hotel bills.

Benveniste's results are being widely interpreted as support for homoeopathic medicine. In the light of our investigation, we believe that such use amounts to misuse.

Our visit and investigation were preconditions for the publication of the original article. We acknowledge that we are an oddly constituted group. One of us (J.B.) is a professional magician (and also a MacArthur Foundation fellow) whose presence was originally thought desirable in case the remarkable results reported had been produced by trickery. Another of us (M.J.) has, in the course of a distinguished career, shown the past decade, in studies of errors and inconsistencies in the scientific literature and with the subject of misconduct in science. The third (J.M.) is a journalist with a background in theoretical physics. None of us has first-hand experience in the field of work at INSERM 200.

We acknowledge that we might well have found ourselves unable to get to grips with the work of the laboratory. But, on the basis of our experience, we are confident that the work of Dr Benveniste and reported by INSERM 200 is inadequate as a basis for the claims made last month and that the defects we shall catalogue are a sufficient explanation of the remarkable results then reported.

We believe that experimental data have been uncritically assessed and their imperfections inadequately reported. We found that Dr Benveniste was misled and then cherished a delusion about the interpretation of its data.

We are grateful to Dr Jacques Benveniste for his openness in discussing some of the questions we raised with him. He allowed us to borrow and to photocopy the relevant laboratory notebooks, which were invaluable for our investigation. We also thank him for the hospitality that Benveniste was (and, perhaps, still is) convinced of the reality of the phenomena reported in his article. We are also in the debt of several of Dr Benveniste's colleagues, especially to Dr Elisabeth Davenas. On her fell most of the burden of demonstrating the standard dilution experiments and of repeating them in a blinded protocol under our scrutiny. We know that our report will be a disappointment to her, and we regret this.

What follows is a narrative account of our visit and a summary of our conclusions.

Our investigations concentrated exclusively on the experimental system on which the publication was based. During our week in Paris, we resisted several proffered opportunities to examine other systems in which high dilution is claimed not to diminish the biological effectiveness of a molecule.

The experimental system has evolved from a test for assessing the susceptibility of people to specific allergens. The guiding principle is that blood-borne allergens have the specific effect of interacting with IgE molecules on the surface of basophils, causing them to degranulate — that is, to release the contents of cytoplasmic granules carrying histamine (and other active substances provoking the symptoms of asthma and hay-fever).

These allergic reactions are apparently mediated at least in part by IgE molecules attached to the surfaces of basophils (in the blood) and mast cells (in tissues). Normally, degranulation is triggered by the interaction of an altered IgE molecule with an antigen, but the same effect can be brought about by the use of anti-IgE — antibody prepared by injecting human IgE into an animal of another species. (INSERM 200 uses goat anti-IgE) at a concentration of 1 mg cm^{-3} sold by the Dutch company Nordic.

The laboratory notebooks provide ample evidence that this expected degranulation is a maximum between log(dilution) 2 and 4.

Benveniste described the published procedure as a "simple experiment". A buffered solution of anti-IgE is serially diluted by a factor of 10 by transferring measured volumes from one test-tube to another. Pipette tips are discarded after each transfer. Basophils are prepared by suspended white cells drained from human blood are transferred to wells in a polystyrene plate. To each of these is added a measured volume of serially diluted anti-IgE or buffer as a control. The wells are incubated for 30 minutes at 37°C. An acidic solution of toluidine blue, which stains intact but not degranulated basophils, is added, and the percentage of recognizable basophils counted via a haemocytometer slide. Anti-IgG, which does not degranulate basophils, is used as a control.

We were surprised to learn that the experiments do not always "work". There have been periods of several months at a time during which solutions at high dilution have failed to degranulate basophils. Indeed, the laboratory had just emerged from such a period. (Speculation at the laboratory is that the distilled water may have been contaminated, or otherwise made unsuitable.) It also appears that

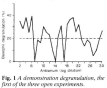


Fig. 1 *A demonstration degranulation, the first of the three open experiments.*

© 1988 Nature Publishing Group

Nature Vol. 334, **28 July 1988**

The investigators concluded:

> *"We conclude that there is no substantial basis for the claim that anti-IgE at high dilution retains its biological effectiveness."*

What went wrong?

- **No serious blinding** — the person counting cells knew which tubes were "active"

What went wrong?

- **No serious blinding** — the person counting cells knew which tubes were "active"
- **Observer bias** — samples that did not "work" were excluded or re-run

What went wrong?

- **No serious blinding** — the person counting cells knew which tubes were "active"
- **Observer bias** — samples that did not "work" were excluded or re-run
- **Data errors of the magnitude that would be expected** from sampling noise

What went wrong?

- **No serious blinding** — the person counting cells knew which tubes were "active"
- **Observer bias** — samples that did not "work" were excluded or re-run
- **Data errors of the magnitude that would be expected** from sampling noise
- **Confirmation bias** — the word "working" was used only for positive results

What went wrong?

- **No serious blinding** — the person counting cells knew which tubes were "active"
- **Observer bias** — samples that did not "work" were excluded or re-run
- **Data errors of the magnitude that would be expected** from sampling noise
- **Confirmation bias** — the word "working" was used only for positive results
- **Financial conflict of interest** — two co-authors were paid by a homeopathic company

The lesson from Benveniste

Extraordinary claims require extraordinary evidence

Blinding, independent replication, and rigorous documentation are not bureaucratic obstacles — they are the defences against our own wishful thinking.

Benveniste was probably not lying — he genuinely believed his results.

That is precisely what makes this case so instructive.

Section 2

Case 2: The Potti Scandal

The promise: personalised cancer therapy

Starting in **2006**, Anil Potti and colleagues at **Duke University** published a series of high-profile papers.

The claim: microarray **gene expression signatures** from cancer cell lines can predict which drug a patient will respond to.

Patients were being **enrolled in clinical trials** and allocated to treatment arms on this basis.



The Annals of Applied Statistics
2009, Vol. 3, No. 3, 1309–1334
DOI: 10.1214/09-AOAS291
© Institute of Mathematical Statistics, 2009

DERIVING CHEMOSENSITIVITY FROM CELL LINES: FORENSIC BIOINFORMATICS AND REPRODUCIBLE RESEARCH IN HIGH-THROUGHPUT BIOLOGY

BY KEITH A. BAGGERLY[1] AND KEVIN R. COOMBES[2]

University of Texas

High-throughput biological assays such as microarrays let us ask very detailed questions about how diseases operate, and promise to let us personalize therapy. Data processing, however, is often not described well enough to allow for exact reproduction of the results, leading to exercises in "forensic bioinformatics" where aspects of raw data and reported results are used to infer what methods must have been employed. Unfortunately, poor documentation can shift from an inconvenience to an active danger when it obscures not just methods but errors. In this report we examine several related papers purporting to use microarray-based signatures of drug sensitivity derived from cell lines to predict patient response. Patients in clinical trials are currently being allocated to treatment arms on the basis of these results. However, we show in five case studies that the results incorporate several simple errors that may be putting patients at risk. One theme that emerges is that the most common errors are simple (e.g., row or column offsets); conversely, it is our experience that the most simple errors are common. We then discuss steps we are taking to avoid such errors in our own investigations.

1. Background.

1.1. *General.* High-throughput biological assays such as microarrays let us ask very detailed questions about how diseases operate, and promise to let us personalize therapy. For example, if we know a priori that only about 70% of a group of cancer patients will respond to standard front-line therapy (as is the case with ovarian cancer), then an array-based test indicating whether a given patient is likely to respond would have great clinical utility.

Data processing in high-throughput studies, however, is often not described well enough to allow for exact reproduction of the results. While various groups have made efforts to construct tools or compendia that should make such reproducibility easier [e.g., Leisch (2002), Ruschhaupt et al. (2004), Gentleman (2005), Gentleman and Temple Lang (2007), Li (2008)], a recent survey [Ioannidis et al. (2009)] of 18 quantitative papers published in *Nature Genetics* in the past two years

Received September 2009.
[1]Supported in part by NIH Grant P50 CA083639 09.
[2]Supported in part by NIH Grant P50 CA070907 08.
Key words and phrases. Microarrays, reproducibility, forensic bioinformatics.

The forensic bioinformatics

Keith Baggerly and Kevin Coombes at **MD Anderson** tried to reproduce the results as collaborators.

The forensic bioinformatics

Keith Baggerly and Kevin Coombes at **MD Anderson** tried to reproduce the results as collaborators.

They could not. So they did something remarkable: they reverse-engineered the analyses from the published figures.

The forensic bioinformatics

Keith Baggerly and Kevin Coombes at **MD Anderson** tried to reproduce the results as collaborators.

They could not. So they did something remarkable: they reverse-engineered the analyses from the published figures.

> *"Data processing is often not described well enough to allow for exact reproduction of the results, leading to exercises in **forensic bioinformatics**."*
> — Baggerly & Coombes, *Ann. Appl. Stat.* 2009

What they found

The errors were **remarkably simple**:

- **Off-by-one row/column shifts** when merging data tables

What they found

The errors were **remarkably simple**:

- **Off-by-one row/column shifts** when merging data tables
- **Swapped sensitive/resistant labels** — the signature would treat patients with the *opposite* drug

What they found

The errors were **remarkably simple**:

- **Off-by-one row/column shifts** when merging data tables
- **Swapped sensitive/resistant labels** — the signature would treat patients with the *opposite* drug
- **Wrong dataset** used for validation — an acute leukemia dataset used to validate a solid tumor predictor

What they found

The errors were **remarkably simple**:

- **Off-by-one row/column shifts** when merging data tables
- **Swapped sensitive/resistant labels** — the signature would treat patients with the *opposite* drug
- **Wrong dataset** used for validation — an acute leukemia dataset used to validate a solid tumor predictor
- **Reuse of training data** as "independent" test sets

What they found

The errors were **remarkably simple**:

- **Off-by-one row/column shifts** when merging data tables
- **Swapped sensitive/resistant labels** — the signature would treat patients with the *opposite* drug
- **Wrong dataset** used for validation — an acute leukemia dataset used to validate a solid tumor predictor
- **Reuse of training data** as "independent" test sets

What they found

The errors were **remarkably simple**:

- **Off-by-one row/column shifts** when merging data tables
- **Swapped sensitive/resistant labels** — the signature would treat patients with the *opposite* drug
- **Wrong dataset** used for validation — an acute leukemia dataset used to validate a solid tumor predictor
- **Reuse of training data** as "independent" test sets

The most common errors are simple errors.

The consequences

- Three **clinical trials** were placed on hold (2010)

The consequences

- Three **clinical trials** were placed on hold (2010)
- Multiple papers **retracted** from high-profile journals

The consequences

- Three **clinical trials** were placed on hold (2010)
- Multiple papers **retracted** from high-profile journals
- Potti **resigned** from Duke

The consequences

- Three **clinical trials** were placed on hold (2010)
- Multiple papers **retracted** from high-profile journals
- Potti **resigned** from Duke
- Duke University paid **$10 million** to settle lawsuits from patients

The consequences

- Three **clinical trials** were placed on hold (2010)
- Multiple papers **retracted** from high-profile journals
- Potti **resigned** from Duke
- Duke University paid **$10 million** to settle lawsuits from patients
- A **US Senate investigation** was launched into clinical trial oversight

The consequences

- Three **clinical trials** were placed on hold (2010)
- Multiple papers **retracted** from high-profile journals
- Potti **resigned** from Duke
- Duke University paid **$10 million** to settle lawsuits from patients
- A **US Senate investigation** was launched into clinical trial oversight

The consequences

- Three **clinical trials** were placed on hold (2010)
- Multiple papers **retracted** from high-profile journals
- Potti **resigned** from Duke
- Duke University paid **$10 million** to settle lawsuits from patients
- A **US Senate investigation** was launched into clinical trial oversight

This was not fraud in the traditional sense

The errors were "simple" — but they were never caught because the raw data and analysis code were never shared.

The lesson from Potti

Reproducible research is a patient safety issue.

- Share raw data — not just summary tables

The lesson from Potti

Reproducible research is a patient safety issue.

- Share raw data — not just summary tables
- Share analysis code — not just methods paragraphs

The lesson from Potti

Reproducible research is a patient safety issue.

- Share raw data — not just summary tables
- Share analysis code — not just methods paragraphs
- **Independent reanalysis** should be possible and routine

The lesson from Potti

Reproducible research is a patient safety issue.

- Share raw data — not just summary tables
- Share analysis code — not just methods paragraphs
- **Independent reanalysis** should be possible and routine
- High-throughput data creates a gap between complexity and intuition that makes errors nearly invisible

The lesson from Potti

Reproducible research is a patient safety issue.

- Share raw data — not just summary tables
- Share analysis code — not just methods paragraphs
- **Independent reanalysis** should be possible and routine
- High-throughput data creates a gap between complexity and intuition that makes errors nearly invisible

The lesson from Potti

Reproducible research is a patient safety issue.

- Share raw data — not just summary tables
- Share analysis code — not just methods paragraphs
- **Independent reanalysis** should be possible and routine
- High-throughput data creates a gap between complexity and intuition that makes errors nearly invisible
  "Poor documentation can shift from an inconvenience to an active danger when it obscures not just methods but errors." — Baggerly & Coombes

Section 3

Case 3: Chocolate and Nobel Prizes

Published in the New England Journal of Medicine

In **October 2012**, Franz Messerli published in *NEJM* (one of the most prestigious medical journals):

"Chocolate Consumption, Cognitive Function, and Nobel Laureates"

The logic: flavonoids in chocolate improve cognitive function → countries that eat more chocolate should have more Nobel laureates.

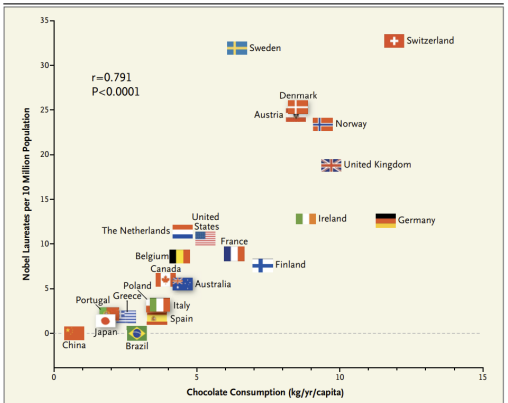


Figure 1. **Correlation between Countries' Annual Per Capita Chocolate Consumption and the Number of Nobel Laureates per 10 Million Population.**

The statistics

The correlation between per-capita chocolate consumption and Nobel laureates per 10 million population across 23 countries:

$$r = 0.791, \quad P < 0.0001$$

The statistics

The correlation between per-capita chocolate consumption and Nobel laureates per 10 million population across 23 countries:

$$r = 0.791, \quad P < 0.0001$$

Highly statistically significant.

The statistics

The correlation between per-capita chocolate consumption and Nobel laureates per 10 million population across 23 countries:

$$r = 0.791, \quad P < 0.0001$$

Highly statistically significant.

To increase a country's Nobel laureate count by **1**, it would need to increase chocolate consumption by **0.4 kg per capita per year**.

The statistics

The correlation between per-capita chocolate consumption and Nobel laureates per 10 million population across 23 countries:

$$r = 0.791, \quad P < 0.0001$$

Highly statistically significant.

To increase a country's Nobel laureate count by **1**, it would need to increase chocolate consumption by **0.4 kg per capita per year**.

The paper **concludes**: *"Chocolate consumption enhances cognitive function, which is a sine qua non for winning the Nobel Prize."*

What is wrong here?

- **Correlation** $\neq$ **causation** — a fundamental principle

What is wrong here?

- **Correlation** $\neq$ **causation** — a fundamental principle
- **Ecological fallacy** — country-level averages say nothing about individuals

What is wrong here?

- **Correlation $\neq$ causation** — a fundamental principle
- **Ecological fallacy** — country-level averages say nothing about individuals
- **Reverse causation** — wealthy, highly-educated countries both consume more chocolate *and* produce more Nobel laureates

What is wrong here?

- **Correlation $\neq$ causation** — a fundamental principle
- **Ecological fallacy** — country-level averages say nothing about individuals
- **Reverse causation** — wealthy, highly-educated countries both consume more chocolate *and* produce more Nobel laureates
- **Confounding** — GDP, research funding, education, climate... all correlated with both variables

What is wrong here?

- **Correlation $\neq$ causation** — a fundamental principle
- **Ecological fallacy** — country-level averages say nothing about individuals
- **Reverse causation** — wealthy, highly-educated countries both consume more chocolate *and* produce more Nobel laureates
- **Confounding** — GDP, research funding, education, climate... all correlated with both variables
- **No mechanism established** — the flavonoid hypothesis is extrapolated far beyond the evidence

Why did it get published?

Why did it get published?

The paper was submitted as **"Occasional Notes"** — a lighter format in NEJM.

Why did it get published?

The paper was submitted as **"Occasional Notes"** — a lighter format in NEJM.

The author later acknowledged it was partly tongue-in-cheek.

Why did it get published?

The paper was submitted as **"Occasional Notes"** — a lighter format in NEJM.

The author later acknowledged it was partly tongue-in-cheek.

But the paper has been cited **hundreds of times**, often uncritically, as evidence for dietary effects on brain function.

Why did it get published?

The paper was submitted as **"Occasional Notes"** — a lighter format in NEJM.

The author later acknowledged it was partly tongue-in-cheek.

But the paper has been cited **hundreds of times**, often uncritically, as evidence for dietary effects on brain function.

The lesson

the prestige of a journal does not guarantee the validity of a finding. **Critical reading is always required.**

The lesson from chocolate & Nobel prizes

Statistically significant $\neq$ scientifically meaningful.

- Observational studies cannot establish causation

The lesson from chocolate & Nobel prizes

Statistically significant $\neq$ scientifically meaningful.

- Observational studies cannot establish causation
- A p-value tells you the probability of the data given no effect — it does not tell you **why** the effect exists

The lesson from chocolate & Nobel prizes

Statistically significant $\neq$ scientifically meaningful.

- Observational studies cannot establish causation
- A p-value tells you the probability of the data given no effect — it does not tell you **why** the effect exists
- With enough variables, spurious correlations are inevitable

The lesson from chocolate & Nobel prizes

Statistically significant $\neq$ scientifically meaningful.

- Observational studies cannot establish causation
- A p-value tells you the probability of the data given no effect — it does not tell you **why** the effect exists
- With enough variables, spurious correlations are inevitable
- Always ask: *What is the plausible mechanism? What are the confounders?*

Section 4

Three Cases, Common Threads

What connects them?

Case	Main failure
Benveniste	Confirmation bias, lack of blinding
Potti	Undocumented analysis, no data sharing
Chocolate/Nobel	Confounding, correlation $\neq$ causation

What connects them?

Case	Main failure
Benveniste	Confirmation bias, lack of blinding
Potti	Undocumented analysis, no data sharing
Chocolate/Nobel	Confounding, correlation $\neq$ causation

All three were published in top journals.

What connects them?

Case	Main failure
Benveniste	Confirmation bias, lack of blinding
Potti	Undocumented analysis, no data sharing
Chocolate/Nobel	Confounding, correlation $\neq$ causation

All three were published in top journals.

All three were eventually corrected — but only after **harm was done**.

The path forward

- **Pre-register** your hypotheses before collecting data

The path forward

- **Pre-register** your hypotheses before collecting data
- **Share** raw data and analysis code (open science)

The path forward

- **Pre-register** your hypotheses before collecting data
- **Share** raw data and analysis code (open science)
- **Blind** your analyses where possible

The path forward

- **Pre-register** your hypotheses before collecting data
- **Share** raw data and analysis code (open science)
- **Blind** your analyses where possible
- **Report effect sizes**, not just p-values

The path forward

- **Pre-register** your hypotheses before collecting data
- **Share** raw data and analysis code (open science)
- **Blind** your analyses where possible
- **Report effect sizes**, not just p-values
- Embrace **reproducible workflows** (R Markdown, Jupyter, ...)

The path forward

- **Pre-register** your hypotheses before collecting data
- **Share** raw data and analysis code (open science)
- **Blind** your analyses where possible
- **Report effect sizes**, not just p-values
- Embrace **reproducible workflows** (R Markdown, Jupyter, ...)
- **Replicate** before acting on findings, especially high-stakes ones

Take-home message

Reproducibility is not a bureaucratic burden.

It is the mechanism by which science corrects itself — and the guarantee that our findings are worth building on.

Take-home message

Reproducibility is not a bureaucratic burden.

It is the mechanism by which science corrects itself — and the guarantee that our findings are worth building on.

In metabolomics, where data matrices are large, analysis pipelines are complex, and the pressure to find biomarkers is high, these principles are not optional.